(a) Candidate Pool Budget K & Latency Trade-off

Stage-1 Candidate Recall@K (%)

Candidate Retrieval Latency (ms)

Candidate Pool Budget (K)

Pareto Knee ($K = 50$)
[CA Recall: 60.1%]

CA (Sparse Regional)
NYC (Compact Dense)
TKY (Dense Network)
Latency (ms/sample)

(b) RRF Smoothing Parameter μ across City Densities

RRF Fused Top-10 Recall (%)

RRF Smoothing Constant (μ)

Robust Plateau: $\mu \in [15, 60]$
(Max variation < 0.8% across all 3 cities)

CA (Sparse Regional)
NYC (Compact Dense)
TKY (Dense Network)

(c) Semantic Cluster Number K_c Trade-off

End-to-End HR@10 (%)

Non-singleton Bucket Coverage (%)

Semantic Cluster Vocabulary Capacity (K_c)

Optimal $K_c = 256$
(CA: 42.5%, NYC: 58.1%)

CA (Sparse Regional)
NYC (Compact Dense)
TKY (Dense Network)
Bucket Coverage (%)

(d) Profile Token Length L_{prof} Trade-off

End-to-End Retention HR@10 (%)

Valid JSON Parse Rate (%)

Max Profile Token Length (L_{prof})

Optimal $L_{prof} = 120$
(100% Parse, CA: 42.5%)

CA (Sparse Regional)
NYC (Compact Dense)
TKY (Dense Network)
Valid JSON Parse Rate (%)